Normal Calculations with a Z-Table KEY

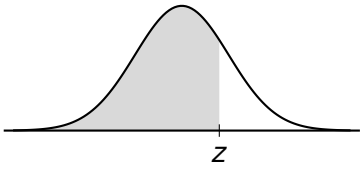
Name _____

The empirical rule only works for whole SDs (1, 2, or 3). A **z-table** works for any z-score.

The **z-table** gives the area to the **left** of z. That is the proportion of the data **below** z.

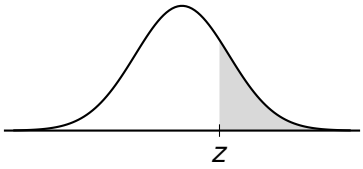
Reading the table: Find $z = 1.33$. Go down to the row **1.3**. Go across to the column **.03**. The area is **.9082**.

Below z



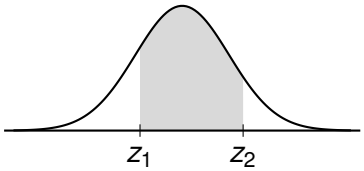
use the table

Above z



$1 -$ table

Between

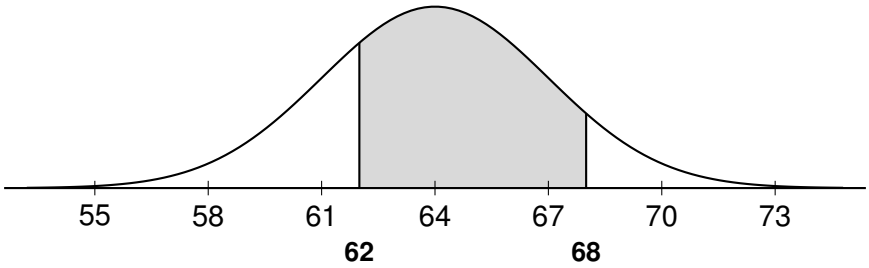


table(z_2) $-$ table(z_1)

Steps

1. Draw the curve. Shade the part the question asks about.
2. Find each z-score: $z = \dfrac{\text{value} - \text{mean}}{\text{SD}}$. Round to 2 decimal places.
3. Look up each z-score in the table.
4. Below: use the table. Above: 1$-$ table. Between: subtract.

Example: Heights of adult women: mean 64 inches, SD 3 inches.



Height of adult women (inches)

Shorter than 68 inches	$z = \frac{68-64}{3} = 1.33$ table: .9082 **90.82%**
Taller than 68 inches	$1 - .9082 = .0918$ **9.18%**
From 62 to 68 inches	$z = \frac{62-64}{3} = -0.67$ table: .2514 $.9082 - .2514 = .6568$ **65.68%**

Use your z-table sheet for the problems.

Work with a partner. Compare answers when you finish each problem.

1. Use the z-table. Find the area **below** each z-score.

   (a) $z = 1.25$ .8944

   (b) $z = -0.84$ .2005

   (c) $z = 2.00$ .9772

   (d) $z = -1.96$ .0250

2. Find the area **above** each z-score.

   (a) $z = 0.50$ $1 - .6915 = .3085$

   (b) $z = -1.20$ $1 - .1151 = .8849$

3. Find the area **between** the two z-scores.

   (a) $z = -1$ and $z = 1$ $.8413 - .1587 = .6826$

   (b) $z = 0.5$ and $z = 1.5$ $.9332 - .6915 = .2417$

4. The empirical rule says about 68% of the data is within 1 SD of the mean. Your answer to 3(a) is .6826. Do these agree? Explain.

   Yes. .6826 is 68.26%. The empirical rule rounds it to 68%.

5. Commute times to a school have a normal distribution with a mean of 25 minutes and an SD of 5 minutes. What percent of students have a commute:

   (a) less than 30 minutes?
   $z = \frac{30-25}{5} = 1$. Table: .8413. 84.13%

   (b) more than 33 minutes?
   $z = \frac{33-25}{5} = 1.6$. $1 - .9452 = .0548$. 5.48%

   (c) from 20 to 28 minutes?
   $z = -1$: .1587. $z = 0.6$: .7257. $.7257 - .1587 = .5670$. 56.70%

6. Apples from an orchard have a mean weight of 150 grams and an SD of 10 grams. The weights have a normal distribution.

   (a) The store sells apples over 165 grams as "extra large." What percent of apples are extra large?
   $z = \frac{165-150}{10} = 1.5$. $1 - .9332 = .0668$. 6.68%

   (b) Apples under 138 grams are used for juice. What percent go to juice?
   $z = \frac{138-150}{10} = -1.2$. Table: .1151. 11.51%